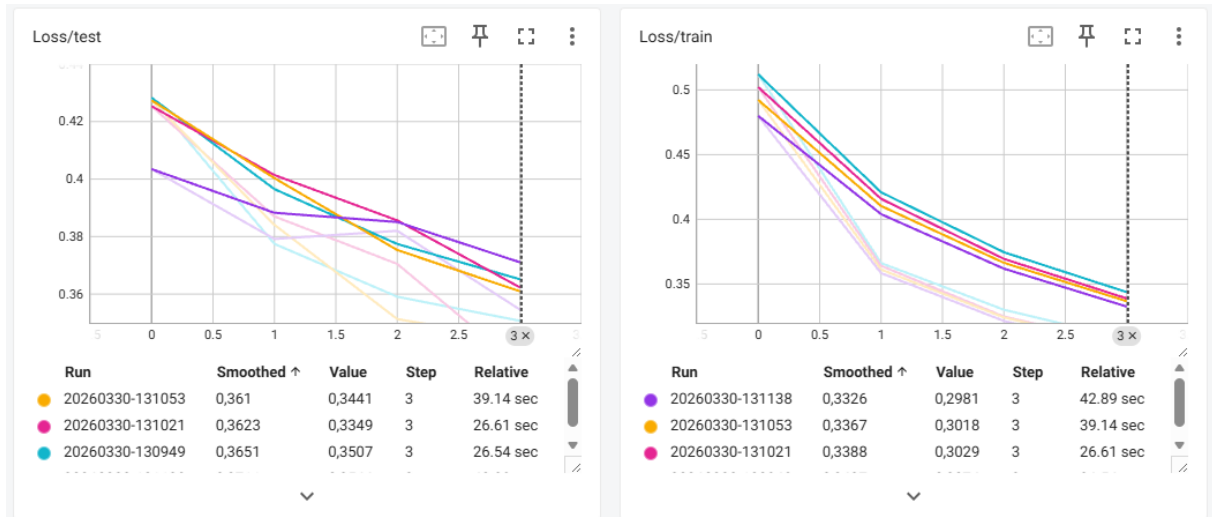


Since I will be applying some more changes in my model, I will start with a baseline run that is a bit faster than my last script. So I adjusted units to 256 and 512 and amount of epochs to 4.

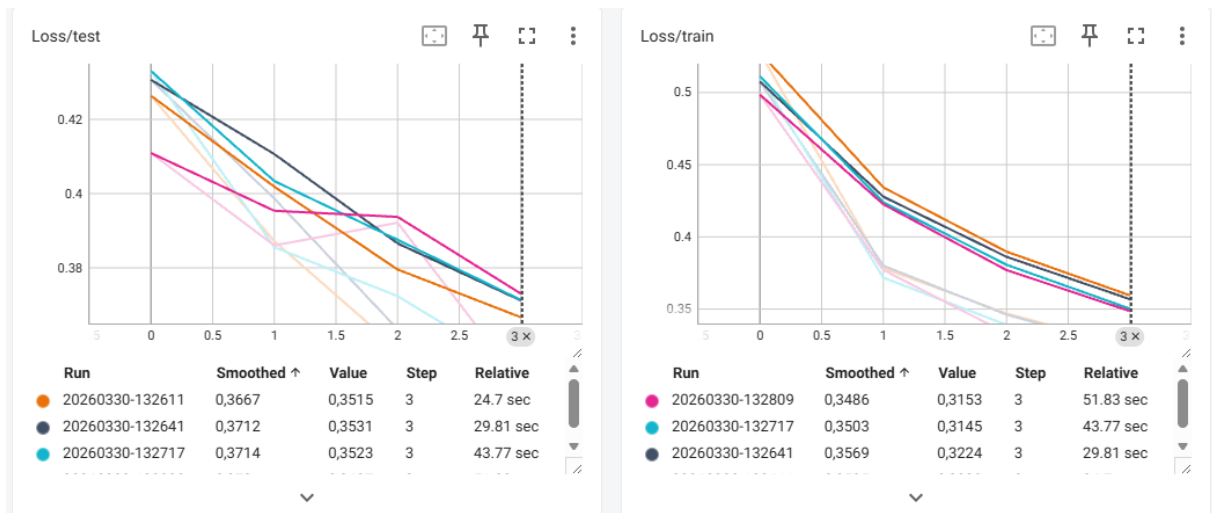
Baseline run:



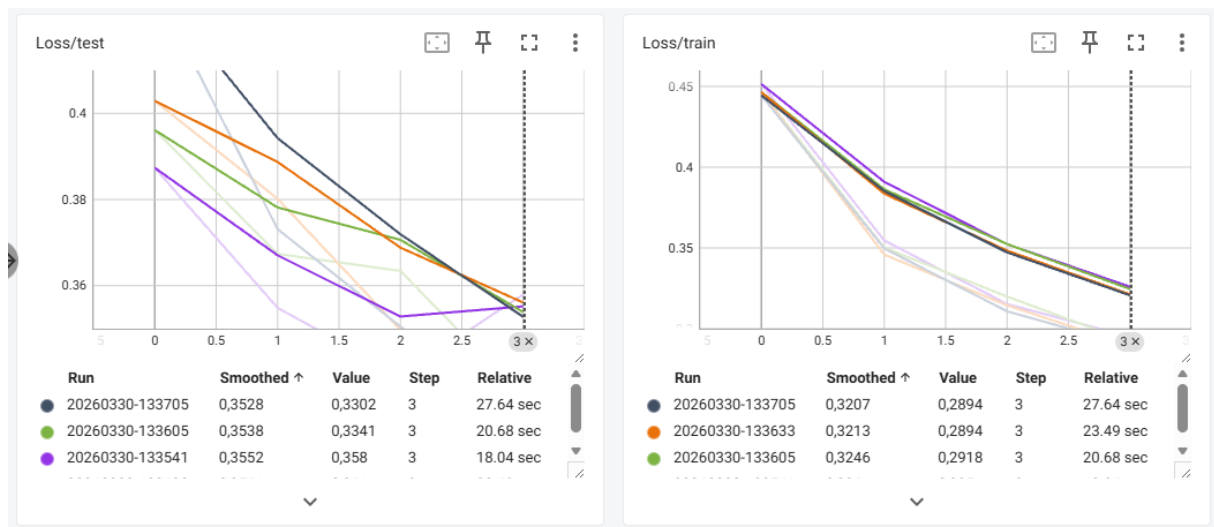
I will start with a hypothesis again.

After adding dropout, normalization, extra convolution layer and extra pooling layer, I expect the training loss and test loss to decrease, while I expect the metric value to improve. At the very least, I expect the generalization of the model to improve (especially after adding dropout).

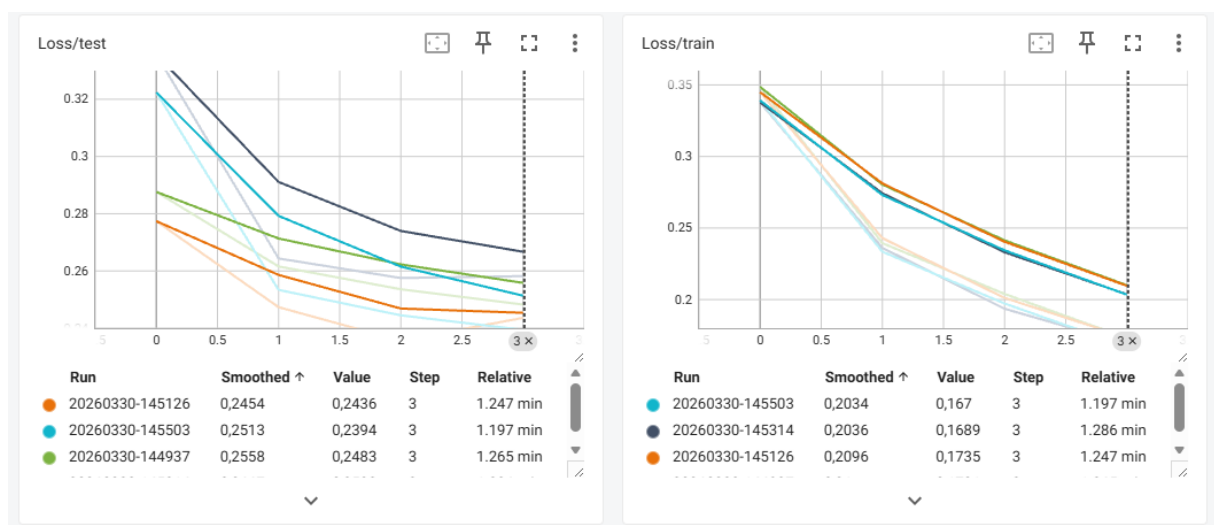
My next run included dropout added of 0.1 after 2 layers. Results actually have gotten a bit worse, but I can imagine this happening since we are leaving out more information, hence the model will have a bit of a harder time to capture certain patterns (especially since I lowered the amount of epochs).



After including dropout, I added normalization and compared the results again. While still including dropout. So now we can see the results are actually improving beyond the baseline run.



Now, convolutional and pooling added, which delivered an incredible improvement on both the training and test loss.



I did all of this using Tensor Board, but now I also added logic to log via MLFLOW and reran the latest run (including convolutional and pooling), so I can show that logging works via MLFLOW.

mlflow 3.5.1

Experiments > cnn_fashion_experiment Machine learning

Search: metrics.rmse < 1 and params.model = "tree"

Time created: State: Active Datasets Sort: Created Columns + New run

	Run Name	Created	Duration	Metrics			Parameters		
				Loss/test	Loss/train	learning_rate	metric/Accuracy	units1	units2
☐	● persistent-mule-2	18 minutes ago	2.5min	0.24254630544246772	0.17563113570213318	0.001	0.9181690705128205	512	512
☐	● silent-pug-7	20 minutes ago	2.2min	0.2617569127573799	0.17150720953941345	0.001	0.9091546474358975	512	256
☐	● amusing-croc-218	22 minutes ago	2.2min	0.2457043036148624	0.17922183871269226	0.001	0.9174679487179487	256	512
☐	● fun-elk-775	24 minutes ago	2.1min	0.2786041274905587	0.173301562666893	0.001	0.9056490384615384	256	256

